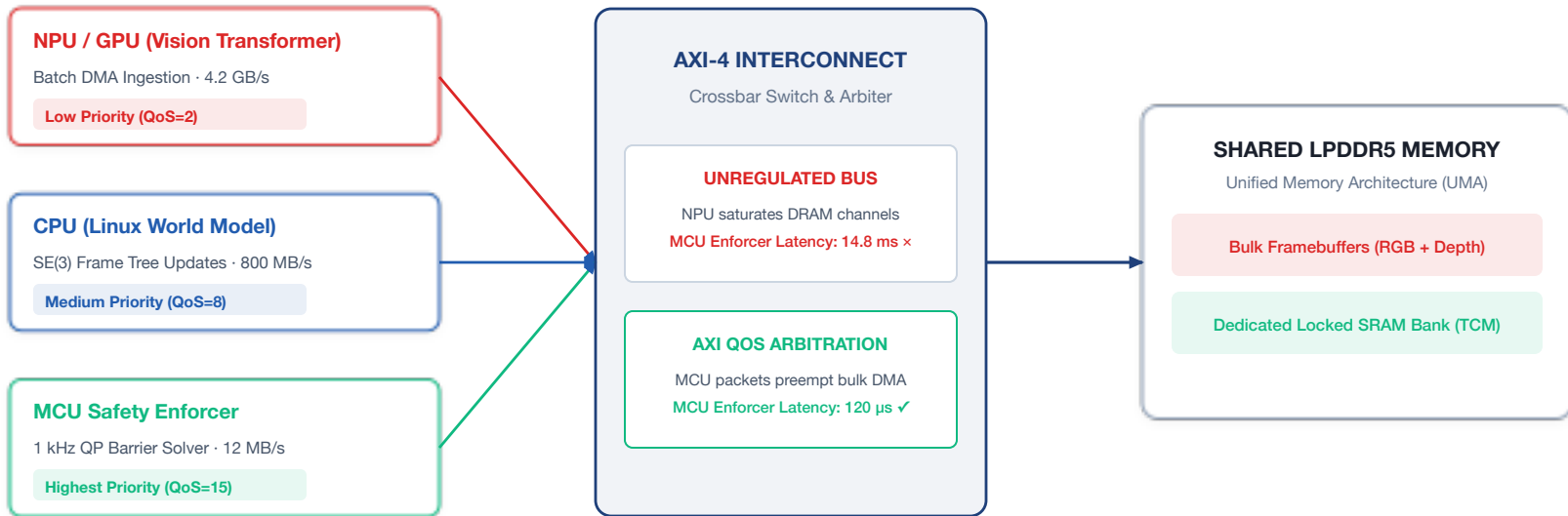


UMA SHARED MEMORY CONTENTION & AXI QoS ARBITRATION

Preventing Tail-Latency Spikes When Vision Ingestion Saturates the Memory Crossbar



THE UMA MEMORY INVARIANT

Real-time safety loops must never share un-arbitrated memory buses with high-bandwidth sensory ingestion pipelines without hardware QoS rate limiting.